THE DISPATCH · NEWSLETTER INTELLIGENCE ENGINE

Components Map

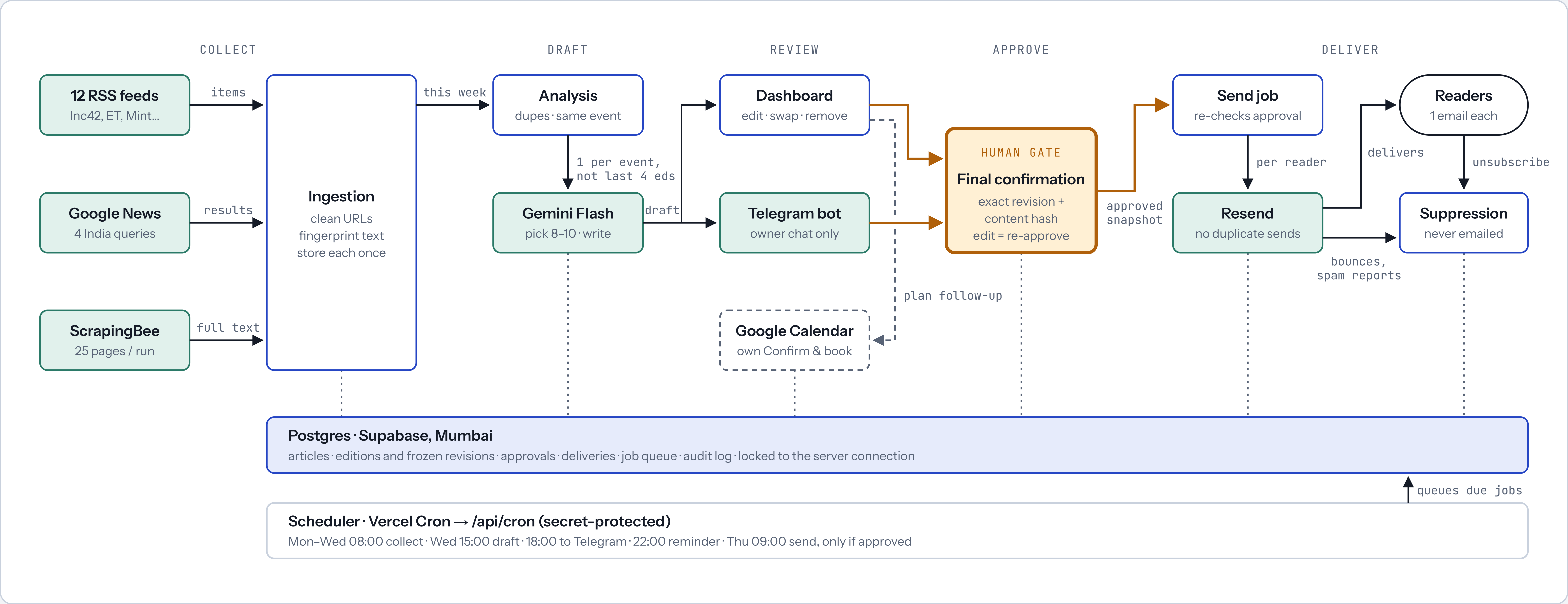
How a week of Indian startup news becomes one approved email, and which parts are allowed to reach the outside world.

LIVE the-dispatch-liard.vercel.app   CODE github.com/aditshah-sketch/dispatch-

The system

Left to right is the week: collect, draft, review, approve, deliver. Everything persists in one Postgres database, and a scheduler queues each step at its time.

▭ Our component ▭ External service ▭ Human approval ⬚ Optional ⋯⋯ Reads and writes the database


COLLECT
DRAFT
REVIEW
APPROVE
DELIVER
12 RSS feeds
Inc42, ET, Mint…
items
Google News
4 India queries
results
ScrapingBee
25 pages / run
full text
Ingestion
clean URLs
fingerprint text
store each once
this week
Analysis
dupes · same event
1 per event,
not last 4 eds
Gemini Flash
pick 8–10 · write
draft
Dashboard
edit · swap · remove
Telegram bot
owner chat only
Google Calendar
own Confirm & book
plan follow-up
HUMAN GATE
Final confirmation
exact revision +
content hash
edit = re-approve
approved
snapshot
Send job
re-checks approval
per reader
Resend
no duplicate sends
delivers
bounces,
spam reports
Readers
1 email each
unsubscribe
Suppression
never emailed
Postgres · Supabase, Mumbai
articles · editions and frozen revisions · approvals · deliveries · job queue · audit log · locked to the server connection
queues due jobs
Scheduler · Vercel Cron → /api/cron (secret-protected)
Mon–Wed 08:00 collect · Wed 15:00 draft · 18:00 to Telegram · 22:00 reminder · Thu 09:00 send, only if approved

The only path to readers runs through the amber gate. The send job verifies, at send time, that the approval matches the current revision and its frozen snapshot, and rechecks each reader against the suppression list before every email.

What can reach the outside world

Every outgoing effect has one thing that authorises it, enforced on the server. Buttons being disabled is never the protection.

EFFECT	GOES THROUGH	AUTHORISED ONLY BY	NEVER AUTHORISED BY
Email to subscribers	Send job → Resend	The owner's final confirmation of the current revision, at or after the chosen send time	An AI draft, story approvals, a reminder, a passed deadline, a calendar event, a background job
Messages to Telegram	Telegram bot	Draft delivery (Wed 18:00), the once-per-edition 22:00 reminder, or the owner's own action	Anyone other than the linked owner chat
Calendar event or invitation	Google Calendar	A separate "Confirm and book" carrying a hash of the exact details reviewed	Approving a story or the newsletter; AI suggestions
Article page fetch	ScrapingBee	A collection run, for thin articles, within 25 pages per run	Paywalled sites and Google redirect links (skipped)
AI processing	Gemini Flash	Draft generation, rewriting one story, suggesting a meeting agenda	Text inside articles, which is treated as data and can't trigger anything

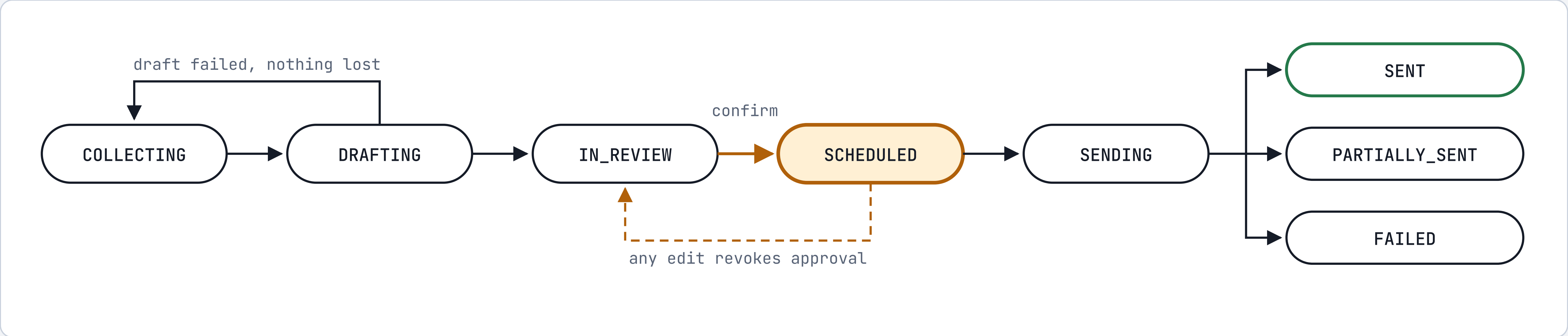
One week, in order

Times are India Standard Time. On Vercel's free plan each step can start up to an hour late.

Mon–Wed · 08:00	**Collect**	Feeds and searches stored once; same-event coverage grouped; last four editions excluded.
Wed · 15:00	**Draft**	Gemini scores candidates, picks 8–10 across 3+ topics and writes 2–3 sentences each.
Wed · 18:00	**Review**	Draft arrives in Telegram with Approve, Edit and Swap on every story.
Wed · 22:00	**Reminder if unconfirmed**	A nudge, once. Silence never sends anything.
Thu · 09:00	**Deliver**	Sends the approved snapshot only. No approval, no email.

Edition states

The edition moves forward only through these steps. Any edit after approval sends it back to review.


draft failed, nothing lost
COLLECTING
DRAFTING
IN_REVIEW
confirm
SCHEDULED
any edit revokes approval
SENDING
SENT
PARTIALLY_SENT
FAILED

An edition counts as published, and joins the four-edition memory, as soon as the first reader's copy is accepted, even if the rest of the send later fails.

Where it runs

One Next.js app on Vercel. Server functions and the database are both in Mumbai, so each page needs only short hops.

PART	RUNS ON	HOLDS OR SENDS
Web app and API	Vercel, region `bom1` (Mumbai)	Public preview at `/preview`; owner desk at `/desk` behind a password and a signed cookie
Database	Supabase Postgres, Mumbai	All workflow state. Row-level security on, API roles revoked, so Supabase's public API can't read it
Scheduler	Vercel Cron → `/api/cron/*`	Six daily triggers; each run recomputes what's due, so duplicate calls are harmless
Secrets	Vercel environment variables	API keys, owner password, signing keys. Never in code, Git history or the browser
Gemini	Google, `gemini-3.8-flash`	Article excerpts in; JSON checked against a schema out
Telegram	Bot `@newsman1213bot`, webhook	Draft cards to the owner; button presses back, verified by a secret header
Resend	Resend email API	One email per reader; delivery, bounce and complaint events back, verified by signature
ScrapingBee	ScrapingBee API	Article URLs out; page text back
Google News	Public search feeds, no key	Search queries out; headlines and publisher names back

Code map

Rules live in one shared layer, so the preview and the live desk enforce the same rules.

LAYER	PATH	RESPONSIBILITY
Domain rules	`src/lib/domain`	Edition commands and approval rules, selection checks, duplicate and same-event detection, history, schedule, email rendering
Preview engine	`src/lib/client/preview-engine.ts`	Runs in the browser on fictional sample data; every outside effect simulated and labelled
Services	`src/lib/server/services`	Ingestion, drafting, editorial, delivery, feedback, Telegram bot, calendar, scheduler
Providers	`src/lib/server/providers`	One adapter per outside service: Gemini, Telegram, Resend, Google, ScrapingBee, feeds
Storage	`src/lib/server/db`, `repo`	Postgres in production, embedded Postgres locally; numbered migrations
HTTP	`src/app/api`, `src/proxy.ts`	Owner session, cron secret, Telegram and Resend webhook checks, one-click unsubscribe

Verified live, 12 October 2026

- **Full journey on production.** Collected news, Gemini draft, Telegram review, owner confirmation, email delivered through Resend, nine stories saved to editorial memory.
- **Collection.** 337 articles from 11 of 12 feeds on production, with 19 full articles via ScrapingBee inside its budget. The 4 Google News queries returned 79 results in a local run; their first production run is the next scheduled collection.
- **Security checks.** Forged Telegram and Resend webhooks rejected; database closed to Supabase's public API; no secrets in Git history.
- **Email reach.** Until a sending domain is verified, Resend delivers only to the account owner's address.
- **Timing.** Vercel's free plan runs scheduled steps once a day, up to an hour late.